

Dexter Fernandes

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Computer Vision Engineer

Machine learning engineer with 3+ years of commercial experience delivering end-to-end ML solutions, from data pipelines through deployment and production monitoring. Skilled in Python, PyTorch, and NumPy, with hands-on experience across supervised learning, time-series forecasting, and model tuning for performance and scalability. Collaborated with cross-functional engineering and client teams to translate performance metrics into deployment decisions.

WORK EXPERIENCE

GrowthStage • Remote

05/2026 - Present

Recruiter

- Evaluated computer vision candidates on their technical depth across model development, deployment, and MLOps practices, maintaining direct exposure to current production engineering standards.

Esbaar • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception in industrial environments, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Built multimodal data ingestion pipelines across image and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records.
- Built an activity classification workstream over 91 sensor channels, using **gradient boosting**, **random forests**, and **MultiRocket** for multivariate time-series classification and forecasting.
- Implemented CI/CD validation workflows for edge inference modules, improving release consistency across hardware targets for 24/7 deployments.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.

LivNSense Technologies • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection models for industrial safety in TensorFlow, reducing false positives by 33% for Austin Industries and John Deere.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.
- Used confusion-matrix failure clustering to diagnose recurring models failure, to prioritise dataset fixes, and improve model robustness across camera feeds.
- Evaluated edge versus cloud inference trade-offs to minimise operational cost while maintaining performance to support ML workloads on Microsoft Azure.
- Deployed quantised vision models using FP16/INT8 optimisation and pruning and layer fusion while meeting production accuracy targets.
- Built automated labelling and versioning workflows, cutting manual labelling effort by 80% while accelerating iteration cycles on large production datasets, reducing model rollout time from 72 hours to under 24.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on a Clearpath Husky, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

PROJECTS

FinanceBench RAG Evaluation Harness

06/2026

- Built a **RAG evaluation harness** for financial-document question answering to diagnose whether failures stem from retrieval, generation, or grounding, rather than reporting a single opaque accuracy figure.
- Exposed runs, results, and failure diagnostics through typed **MCP** tools and a **FastAPI**-based **REST API**, enabling both programmatic and human-readable inspection of the evaluation pipeline.
- Fine-tuned a **bge-reranker-base cross-encoder** with **LoRA** and hard-negative mining for financial-document domain adaptation, self-hosted behind a **containerised inference API**, and benchmarked it against the pretrained model on MRR, nDCG, evidence hit rate, grounding, and downstream answer accuracy, improving retrieval ranking over the pretrained baseline.
- Built a **hybrid-retrieval pipeline** combining **BM25**, **FAISS** dense embeddings, and **reciprocal rank fusion with cross-encoder reranking**, using recursive character chunking and three isolable retrieval modes (closed-book, oracle-context, dense) to separate retrieval quality from generation quality.
- Built a stateful **LangGraph** agent that orchestrates retrieval, correctness, and grounding tools through conditional workflows with multi-step reasoning and structured **Pydantic** outputs, benchmarking diagnostic accuracy, latency, and tool usage against a deterministic baseline.
- Designed a dual-track evaluation layer pairing deterministic checks (exact/normalised match, scale-aware numeric matching) with **LLM-as-judge** scoring, and benchmarked the pipeline against **Snowflake Cortex Search** on identical chunks and metrics to validate retrieval quality independently.

SKILLS

- **Programming & Tooling:** Bash, C++17, CUDA C++, Git, Linux, NumPy, OpenCV, Pandas, PIL, PostgreSQL, Python, PyTorch, Scikit-image, Scikit-Learn, TensorFlow
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, CI/CD, Docker, DVC (Data Version Control), Microsoft Azure, MLflow, Production monitoring, Structured logging
- **Time Series Modelling:** gradient boosting, MultiRocket, random forests, Time-series forecasting
- **Computer Vision Models:** ByteTrack, EfficientDet, Instance Segmentation, NvDCF, NvSort, Object detection, Object tracking, RT-DETR, YOLO
- **Datasets, Annotation & Evaluation:** Confusion-matrix-driven failure clustering, CVAT, Dataset versioning, GRAD-CAM, Synthetic data generation, TIDE (Toolbox for Identifying Detection Errors)